

***What if your app gets hacked
not because of code.. but
because of a leaked secret?***

What are Secrets?

- Sensitive credentials used by applications:
- API keys
- Database passwords
- Access tokens
- Used for authentication & secure communication

Why Secrets are Dangerous?

- Hardcoded secrets = easy target for attackers
- Common risks:
 - GitHub leaks
 - Insider threats
 - Misconfigured access
- One leaked secret → full system compromise

What is Secrets Management?

- Process of securely storing, accessing, and rotating secrets
- Ensures:
- Secrets are not exposed in code
- Access is controlled & monitored
- Centralized and secure handling of credentials

Zero Trust Mindset

- Principle: “Never trust, always verify”
- No system or user is trusted by default
- Every access request must be:
 - Authenticated
 - Authorized
- Applied to secrets:
 - No permanent access
 - No blind trust

Key Practices

- Do not hardcode secrets in code
- Use least privilege access
- Rotate secrets regularly
- Monitor and audit access
- Use environment-based configuration

Real-World Impact

- Many major breaches happen due to:
- Exposed API keys
- Leaked credentials
- Security failure is often not code... but secrets

Key Takeaway

- Secrets are the most sensitive part of your system
- Managing them properly = core of modern security